Day 5: Details of the Combat Talon I Navigation Kalman Filter

- Segment 3: The Combat Talon I Kalman Filter High-Level Software Architecture

- Segment 4: The Combat Talon I Kalman Filter JOVIAL Implementation

Day 5, Segment 3
The High-Level Combat Talon Software Architecture
Topics

- Architecture Overview
- Rate Group Drilldown
- Sequencing

CT I Kalman Filter Software Architecture Overview

- Primary Kalman Cycle: Every 10 sec or .1 Hz
- Secondary Kalman Subcycles: 1 Hz and 4 Hz
- Third Tier Kalman Subcycle: 8 Hz
- Fourth Tier Kalman Minor Cycle: 16 Hz

CT I Kalman Filter Software Architecture Overview

- Highly intertwined with general navigation calculations
- Some of the same utilities are called at various rates

CT I Kalman Filter: Rate Group Drilldown

- Primary Kalman Cycle: Every 10 seconds or .1 Hz
  - Final extrapolation and application of observation update
  - Doppler compensation (weighted averages)
  - Kalman initialization
  - Buffering of Kalman data from 4 Hz and 1 Hz
  - Mag Var initialization
  - Weighted averaging

CT I Kalman Filter: Rate Group Drilldown

- Secondary 1 Hz calculations of ...
  - Earth rate, Earth radii and their inverses
  - TAS, wind, temperature
  - CEP

CT I Kalman Filter: Rate Group Drilldown

- Secondary 1 Hz calculations (cont)
  - Mag Var
  - Gravity
  - Geographic Position Errors
  - GPS Alt Cal deltas and EVE means / standard devs.

CT I Kalman Filter: Rate Group Drilldown

- Secondary 4 Hz subcyle processing performs ...
  - INU Mode Conrol
  - F02 transmission (if in alignment)
  - Transport rate and Coriolis rate calcs
  - Doppler groundtrack and drift angles (vestiges of the now defunct Doppler filter)

CT I Kalman Filter: Rate Group Drilldown

- 4 Hz processing performs (cont)...
  - Direction cosines computation
  - Geographic position and wander angle calcs
  - Position fix and alt cal processing
  - Navigation mode control
  - Observation accumulation
  - Transition matrix accumulation

CT I Kalman Filter: Rate Group Drilldown

- Third Tier Kalman Subcycle: 8 Hz computations for ...
  - Raw Doppler processing
  - Doppler compensation
  - Direction cosine extrapolation
  - Geographic position
  - Non-inertial vertical channel
  - Doppler DR

CT I Kalman Filter: Rate Group Drilldown

- Fourth Tier Kalman Minor Cycle: 16 Hz calculations for ...
  - High-rate INS compensation
  - Geographic position, velocity, altitude
  - Angles: Attitude, heading, groundtrack, angle of attack

CT I Kalman Filter Sequencing

- Process INU 1 filter, then INU 2
- Different processing depending on INU mode: Alignment vs. Navigation

CT I Kalman Filter Sequencing

- Main .1 Hz Sequence:
  - Extrapolate state vector and transition matrix to current time
  - Perform covariance rate limiting and calculate Q diagonals
  - Extrapolate covariance matrix to current time
  - Process observations
    - Gain computation included herein

CT I Kalman Filter Sequencing

- Main .1 Hz Sequence (cont):
  - CEP calcuation
  - Monitor alignment status

Day 5, Segment 5
The JOVIAL Implementation
Topics

- Code Structure
- Extrapolation
- Measurement Update Processing

Combat Talon I Kalman Filter Code Structure

- Separate tasks and real-time execs for each rate group
  - .1 Hz $\Rightarrow$ REP005HZ $\Rightarrow$ KF'NAV'05'HERTZ'EXEC
  - 1 Hz $\Rightarrow$ REP1HZ $\Rightarrow$ KF'NAV'1'HERTZ'EXEC
  - 4 Hz $\Rightarrow$ REP4HZ $\Rightarrow$ KF'NAV'4'HERTZ'EXEC
  - 8 Hz $\Rightarrow$ REP8HZ $\Rightarrow$ KF'NAV'8'HERTZ'EXEC
  - 16 Hz $\Rightarrow$ REP16HZ $\Rightarrow$ KF'NAV'16'HERTZ'EXEC

.1 Hz Kalman Routines

- Was originally .05 Hz
- Basic Kalman update period is 10 secs
- Call Tree:

```
REP005HZ
| KF'NAV'05'HERTZ'EXEC
| | NU'INERTIAL'DOP'DATA'COMP'CORR
| | KF'KALMAN'DATA'BUFFERING
| | NU'MAGNETIC'VAR'SETUP
| | KF'KALMAN'CONTROL
```

.1 Hz Detail

```
REP005HZ
| KF'NAV'05'HERTZ'EXEC
| | NU'INERTIAL'DOP'DATA'COMP'CORR
| | KF'KALMAN'DATA'BUFFERING
| | | KF'BEGIN'INU'INIT
| | | KF'AIDED'INS'FILTER'INIT
| | | KF'EARTH'RAD'BUFFER
| | | KF'BEGIN'BUF'MOV'POS
| | NU'MAGNETIC'VAR'SETUP
| | | NU'MAGNETIC'VAR'CALC
```

.1 Hz Detail

```
| | KF'KALMAN'CONTROL
| | | KF'AVERAGE'WEIGHT'CAL
| | | KF'EXTRAP'ERROR'STATE'VECTOR
| | | KF'TRANS'MATRIX'CALCULATION *
| | | KF'COVARIANCE'RATE'LIMIT
| | | KF'COV'EXTRAPOLATION
| | | KF'UPDATE'PROCESSING'CONTROL
| | | KF'POSITION'FIX'PREP
| | | KF'INU'COVARIANCE'CAL
| | | KF'CEP'CALCULATION
| | | KF'MOTION'ALIGN'STATUS
```

Extrapolation

- System Dynamics and Transition Matrix Structure
- Transition Matrix Accumulation
- State Extrapolation and Covariance Extrapolation

System Dynamics and Transition Matrix Structure

- Block partitioning based on Varying and Non-varying states

- Singer used A for system dynamics matrix and F for state transition matrix

  – instead of usual F for system dynamics and Φ for transition matrix

System Dynamics and Transition Matrix Structure

- Φ block structure:

(1) $$\Phi = \begin{bmatrix} F_{\mathsf{vv}} & F_{\mathsf{nv}} \\ 0 & I \end{bmatrix}$$

where F_{vv} is the block propagating varying states, while F_{nv} propagates non-varying states.

Extrapolation: Transition Matrix Formation and Use

- Transition Matrix Formation: KF'TRANS'MATRIX'CALCULATION
- Sparse Matrix Multiplication: KF'SPARSE'MAT'VECT'MULT
- Let's go code snorkeling!

Covariance Extrapolation

- Covariance Rate Limiting: KF'COVARIANCE'RATE'LIMIT
- Covariance Extaploation: KF'COV'EXTRAPOLATION
- Prepare to snorkel ...

Measurement Update Processing

- Primary Kalman Cycle Measurement Update: KF'UPDATE'PROCESSIN
- Secondary 4 HZ Subcycle: OP'OBSERVATION'PROCESSING
- Prepare to dive ...

Kalman Filter Class Wrap-Up

- Where has our journey taken us?
  - From General to Specific
  - Kalman Filter Theory $\rightarrow$ Navigation Theory $\rightarrow$ Inertial Error Dynamics $\rightarrow$ Combat Talon I
- Acquainted with concepts and terminology
- Acquired some tools for further study

Kalman Filter Class Wrap-Up

- Not an easy subject!
  - mathematically rich
- But not insurmountable!

Kalman Filter Class Wrap-Up

- For further study:

  1. *Kalman Filtering: Theory and Practice Using MATLAB*, M. S. Grewal, A. P. Andrews, Wiley Interscience

  2. *Global Positioning Systems, Inertial Navigation and Integration*, Wiley Interscience

  3. *Introduction to Random Signals and Applied Kalman Filtering with Matlab Exercises and Solutions*, R. G. Brown, P. Y. C. Hwang, Wiley

  4. *Applied Optimal Estimation*, A. Gelb et. al., MIT Press

5. *Integrated Aircraft Navigation*, J. L. Farrell, Academic Press

6. *Strapdown Inertial Naviation Lecture Notes*, (Sixth Printing), Paul G. Savage, Srapdown Associates, Inc.

Kalman Filter Class Wrap-Up

- For Further Study
  - Take Paul Savage's course on strapdown inertial navigation
  - Try to obtain some of the original Singer-Kearfott documentation
  - Play with the Matlab / Simulink toys!
  - Conduct in-house study group seminars on the CT I Kalman filter: Once a week have someone examine and present a detailed analysis and code walkthrough of one of the major KF modules

– Conduct in-house study group seminars comparing the various SOF inertial nav filters: The Talons, the Gunships, the Pave Lows, etc. Learn from each other!

Kalman Filter Class Wrap-Up

- Recommendations for modifying the Talon I filter:
  - Develop a covariance analysis simulation capability
  - Develop a Kalman datapump capability within the MCOFP: log Φ, Q, P, R, K elements

Kalman Filter Class Wrap-Up

- Thanks for being such a good class!
- It's an honor ...
- Thanks to Emory Jiles and SSAI for being such gracious hosts

Kalman Filter Class Wrap-Up

- My contact info:

Michael L. Carroll
mcarroll_91786@yahoo.com
Cell Phone: 925-381-6423

- Thank you and good luck!